

Research on the impact of generative artificial intelligence (GenAI) on enterprise innovation performance: a knowledge management perspective

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Abstract

Purpose – This study aims to investigate the impact of generative artificial intelligence (GenAI) on enterprise innovation performance, particularly from the perspective of knowledge management. It addresses key challenges in GenAI adoption – such as data biases, information overload and technological dependence – and proposes strategies to overcome these obstacles to enhance innovation.

Design/methodology/approach – Adopting a theoretical approach, this research analyzes the role of knowledge management in bridging the gap between GenAI and enterprise innovation. A structured framework based on four essential knowledge management processes – knowledge creation, retrieval and storage, transfer and sharing and application – is developed to tackle these challenges effectively.

Findings – The study reveals that while GenAI presents both opportunities and challenges for enterprise innovation, leveraging a structured knowledge management framework is key to unlocking its potential. It underscores the critical role of human–AI collaboration in mitigating issues such as data biases and integration challenges, ultimately improving innovation performance. The findings highlight the importance of complementing AI capabilities with human judgment to ensure successful outcomes in GenAI-driven innovation.

Research limitations/implications – This conceptual study calls for further empirical research to validate the findings and expand their generalizability. Future studies should explore contextual factors such as organizational characteristics, business environments and policy frameworks to refine the proposed framework.

Originality/value – This research offers novel insights into the intersection of GenAI, knowledge management and enterprise innovation. It stresses the importance of human involvement alongside GenAI, providing actionable recommendations for organizations navigating the complexities of AI adoption. In addition, it contributes to the evolving discourse on AI and innovation management, offering pathways for businesses to harness GenAI's full potential and drive performance.

Keywords Generative artificial intelligence, Knowledge management, Enterprise innovation performance, Human–AI collaboration

Paper type Viewpoint

1. Introduction

As an advanced form of machine learning, generative artificial intelligence (GenAI) systems demonstrate the capability to assimilate existing data patterns and generate novel, contextually relevant outputs across multiple modalities including text, imagery, audio and programming code (Lee and Chung, 2024). This technological breakthrough positions GenAI as a strategic enabler for organizational innovation, evidenced by McKinsey's 2024 AI Adoption Survey revealing a doubling in enterprise deployment rates (33% to 65%) within a single fiscal year. While, such exponential growth exposes critical tensions between

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generative automation and organizational knowledge management systems (KMS), creating both disruptive opportunities and systemic challenges for innovation management.

Enterprises increasingly leverage GenAI tools (e.g. ChatGPT, DALL-E, DeepSeek) to enhance innovation processes across three strategic dimensions:

1. augmenting ideation efficiency through automated content generation;
2. facilitating cross-domain knowledge synthesis via intelligent data analysis; and
3. enabling rapid prototyping through computational creativity (Jia *et al.*, 2024).

These applications demonstrate measurable productivity gains – Microsoft's implementation of GPT-4 reportedly reduced conceptualization-to-prototype cycles by 40% in software development teams. However, the technology's disruptive nature introduces paradoxical challenges: while 68% of surveyed organizations acknowledge GenAI's innovation potential, only 23% have established systematic frameworks for its strategic implementation (Tekic and Tekic, 2024).

This gap becomes particularly evident when examined through the lens of knowledge management, revealing three knowledge-centric barriers that constrain the innovative efficacy of GenAI in organizational contexts (Del Giudice and Della Peruta, 2016). First, the absence of robust knowledge verification mechanisms creates an authenticity gap between GenAI outputs and operational reality. Second, the failure to establish explicit AI-strategy mapping relationships through KMS creates dual challenges: on the one front, AI-generated knowledge fragments resist transformation into innovation solutions; simultaneously, excessive reliance on GenAI outputs precipitates strategic drift. Third, organizational deficiencies in absorptive capacity precipitate three critical challenges: ineffective human–AI collaborative, weak cross-boundary diffusion capabilities of AI-generated innovations and imbalanced human–AI knowledge co-creation patterns stemming.

Literatures reflects these tensions through three theoretical lenses. Proponents emphasize GenAI's combinatorial creativity in expanding organizational idea pools (Lee and Chung, 2024; Shaik *et al.*, 2024), while critics warn of innovation homogenization risks from over-standardized AI outputs (Jia *et al.*, 2024). Emerging evidence reveals a non-linear innovation performance curve, a phenomenon influenced by knowledge management maturity (Guang *et al.*, 2024). This curvilinear relationship underscores the critical yet understudied role of knowledge management practices in translating GenAI knowledge into enterprise innovation capital.

The application of GenAI technology provides significant capabilities for data retrieval and integration, facilitating the generation and dissemination of knowledge. However, only through effective knowledge management can enterprises fully capitalize on the innovative potential of this technology (Santoro *et al.*, 2021). Knowledge management, as a systematic process for managing information and knowledge flow within organizations, fosters the acquisition, sharing and application of knowledge, thereby enhancing innovation capabilities (Abdillah *et al.*, 2024). Understanding GenAI's impact on knowledge management processes is crucial for enterprises to effectively manage and use the knowledge generated, support the creation of new, valuable knowledge and facilitate cross-departmental collaboration, ultimately boosting innovation performance.

Building on previous studies, this research examines the impact of GenAI on enterprise innovation performance from a knowledge management perspective. The structure of the study follows the problem identification, analysis and resolution steps. Section 2 reviews the relevant literature on the topic and identifies existing gaps. Section 3 analyzes issues related to GenAI's empowerment of enterprise innovation performance through knowledge management. Section 4, explores the opportunities and challenges knowledge management faces in leveraging GenAI to drive innovation performance. Section 5 offers trends and

practical insights for the future application of GenAI in enterprises, providing a reference for enterprise management in the AI era.

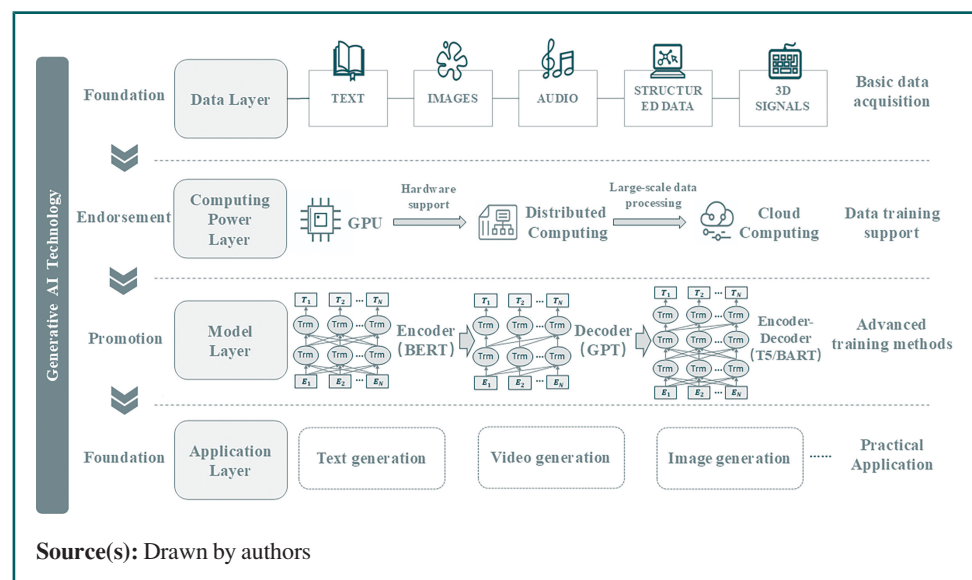
2. Literature review

2.1 Generative artificial intelligence

GenAI refers to AI systems capable of generating high-quality text, images and other content based on training data (Kanbach *et al.*, 2024). Its core lies in pre-training on large volumes of text and other data types, which enables the model to understand, generate and optimize output. As schematically represented in Figure 1, GenAI systems exhibit a four-tiered architecture comprising: data, computation, model and application. The data layer involves large-scale training data sets, data labeling and data preprocessing (Al Naqbi *et al.*, 2024). High-quality data input is essential for the model's ability to generalize, ensuring stability and reliability, which directly influences the performance and effectiveness of GenAI models. The computation layer uses GPUs/TPUs and other hardware to accelerate training and inference for compute-intensive tasks, such as deep learning. Distributed computing enables the effective processing of large-scale data, while cloud computing provides elastic resources that support flexible model scaling. This makes the computation layer a critical material foundation for GenAI. The model layer forms the core of GenAI technology and includes various deep learning models, particularly the transformer architecture. By using self-attention mechanisms and positional encoding, this architecture enables more efficient natural language processing (NLP) and learning (Liang *et al.*, 2024). As model algorithms continue to evolve, they complement traditional models, driving GenAI innovation and application. The model layer thus serves as the technical foundation of GenAI. The application layer is where GenAI technology is implemented, transforming technical results into products and services (Abrokwah-Larbi, 2023).

GenAI elevates enterprise innovation performance through its multi-layered architecture: data diversity stimulates creative insights, computing power accelerates ideation, models facilitate intricate problem-solving and application enable seamless integration of AI-driven innovations into organizational workflows. This paper will further explore the impact of GenAI on enterprise innovation performance from a knowledge management perspective.

Figure 1 Generative AI technology framework



2.2 Generative artificial intelligence and innovation performance

Innovation is commonly defined in the literature as the creation and application of new products, services and organizational processes to meet user needs. GenAI, characterized by its rapid content generation and extensive knowledge coverage, essentially represents the foundational data intelligence derived from human text, video and other media, significantly impacting traditional methods of information acquisition and innovation. The content generated by GenAI spans a broad range of human knowledge domains, offering vast potential for application in enterprise innovation scenarios. Existing research highlights that GenAI holds significant innovation potential, particularly in the following four aspects:

1. Based on the resource-based view, the aggregative and generative features of GenAI enhance enterprises' ability to collect and process new knowledge and information. This is especially valuable for rapid information collection in a fast-changing market, technological transformations, demand forecasting and new product development, thereby strengthening the innovation resources of enterprises. GenAI also creates more opportunities for innovation by generating novel ideas (Wodecki *et al.*, 2019). Research categorizes GenAI resources into three key dimensions: tangible, intangible and human resources (Barney, 1991). These resources are crucial for entrepreneurs implementing GenAI. Tangible resources, such as hardware, data and tools, provide the technical foundation for GenAI applications (Mikalef and Gupta, 2021). Intangible resources, including an innovative culture and strategic initiatives, play a critical role in accelerating the application and innovation of GenAI. Human resources, on the other hand, emphasize the knowledge and capability of personnel to apply GenAI, which is vital for translating technological potential into productive outcomes (Chowdhury *et al.*, 2023).
2. GenAI reorganizes existing knowledge by mining current knowledge, technologies and data within enterprises. Unlike individuals, GenAI transcends cognitive limitations, autonomously learning and iterating while examining resources through the lens of collective knowledge and wisdom. This process enhances the efficiency of enterprise innovation (Ma *et al.*, 2024).
3. From a business process perspective, GenAI effectively liberates employees from simple, repetitive and tedious tasks by reshaping workflows – allowing AI to handle such tasks – thereby enabling employees to focus more on complex, challenging and innovative activities, thus improving enterprise innovation performance (Yin *et al.*, 2024).
4. In terms of innovation management models, GenAI shifts decision-making from traditional experience-based approaches to data-driven innovation. This shift effectively addresses cognitive limitations and biases, unlocking internal innovation potential (Korherr *et al.*, 2022).

In summary, GenAI offers significant opportunities to enhance enterprise innovation performance. However, its empowerment process inevitably faces challenges such as misinformation dissemination, copyright concerns and information overload, which hinder its adoption and integration within enterprises. Therefore, it is essential to further explore the mechanisms through which GenAI influences enterprise innovation performance to better enhance innovation outcomes.

2.3 Knowledge management

Knowledge management is the process of promoting decision-making ability, productivity and creativity through the creation, acquisition, sharing, organization and utilization of knowledge in an appropriate environment, ultimately improving organizational performance (Abubakar *et al.*, 2019). Enterprises can enhance their competitiveness and achieve sustainable competitive advantages through the effective development, acquisition and application of knowledge. Enterprise knowledge management mainly consists of two key

components: the knowledge management environment and the knowledge management process (Maier and Hadrich, 2011). The more critical part is the knowledge management process, as it directly generates and creates knowledge value and exists relatively independently of the formal organizational support system (Ooi et al., 2023).

Different studies define the knowledge management process in various ways, but generally, it includes knowledge acquisition, knowledge storage, knowledge transfer, knowledge sharing and knowledge application (Alavi and Denford, 2012). The operational definition of the knowledge management process is the process of knowledge creation, knowledge sharing and knowledge utilization. The main reason for the differences in definitions lies in the varying levels of aggregation of the processes, though the elements are interrelated (Del Giudice and Maggioni, 2014).

Alavi and Leidner (2001) summarized the knowledge management process framework into four dimensions: knowledge creation, knowledge storage/retrieval, knowledge transfer and sharing and knowledge application. The goal is to create value from the knowledge assets possessed by the enterprise. Knowledge creation refers to developing new knowledge from raw data or combining existing knowledge in new ways; knowledge storage and retrieval involve organizing and maintaining knowledge for future use; knowledge transfer and sharing refer to the circulation of knowledge among multiple entities; and knowledge application means effectively using knowledge for decision-making and problem-solving (Chin et al., 2025).

Knowledge management involves not only the technology of management (machine factors) and the context of management (environmental factors) but also the factors related to multiple stakeholders (human factors), which are considered the three pillars of knowledge management, participating in the entire knowledge management process (Carrion et al., 2004). This paper will explore solution paths for problems encountered in the process of GenAI empowering enterprise innovation performance, based on the four structures of the knowledge management process framework. Existing literature suggests that for knowledge management to be effective, it must address the use and development of technological tools, the design and optimization of knowledge management processes and the human factors and cultural environment at both individual and organizational levels (Alavi and Leidner, 2001; Carrion et al., 2004).

At the technological level, the focus is on how information systems and tools support knowledge acquisition, storage, sharing and application (Becerra-Fernandez and Sabherwal, 2014). At the process level, the emphasis is on the flow and evolution of knowledge within organizations, covering key processes such as creation, retrieval and storage, transfer and sharing and application (Alavi and Leidner, 2001). At the people level, the focus is on organizational members' attitudes, skills and collaborative culture regarding knowledge, as well as behaviors such as technology acceptance or resistance and willingness to share knowledge (Hislop et al., 2018).

It is because these three aspects work together and are closely interconnected in knowledge management that many studies explore the overall framework from the perspectives of technology, processes, and people. This allows for a systematic identification and resolution of the core challenges organizations face when applying and innovating with knowledge. By focusing on these three pillars, researchers and managers can gain a more comprehensive understanding of the opportunities and risks involved in applying emerging technologies (such as GenAI), leading to more effective implementation in management practices and enhanced innovation performance.

2.4 Knowledge management with generative artificial intelligence and innovation performance

The essence of innovation lies in introducing new content and methods continuously to meet users' dynamic and personalized needs. With the deepening of knowledge management

research, the knowledge-based view of innovation emphasizes achieving new products and processes through the creation and application of knowledge (Quintane *et al.*, 2011). The ultimate goal of knowledge management is to achieve innovation, enhance enterprise competitive advantage and ultimately improve organizational performance (Scuotto *et al.*, 2017). Numerous studies have shown that the knowledge management process, practices and infrastructure are closely and positively related to enterprise innovation performance. GenAI influences each stage of the knowledge management process, thereby impacting enterprise innovation performance (Kaczorowska-Spychalska *et al.*, 2024). Specifically, it includes:

- *Knowledge creation*: First, GenAI enhances the acquisition, categorization and absorption of explicit knowledge and also transforms some tacit knowledge into explicit forms through AI methods, providing more material for enterprise innovation (Lichtenthaler, 2009). Second, by systematically integrating fragmented explicit knowledge, identifying patterns and generating insights and suggestions, GenAI strengthens enterprise innovation capabilities (Korzynski *et al.*, 2023).
- *Knowledge retrieval and storage*: First, GenAI significantly broadens the scope of knowledge acquisition, extracting valuable data and information on a global scale, helping enterprises access more heterogeneous information and knowledge. Second, in-depth searches of both internal and external knowledge improve the ability of enterprises to quickly identify valuable information, avoiding information overload and enhancing knowledge retrieval and storage efficiency. Finally, GenAI converts unstructured data into structured knowledge, providing automatic tagging, categorization and indexing for knowledge information, innovatively analyzing content, identifying keywords and assigning appropriate metadata, making it easier to organize and retrieve knowledge during the enterprise innovation process (Kondurkar *et al.*, 2024). It is capable of retrieving and storing knowledge from multiple sources (such as employee inputs, team dynamics, documents and computer files), ensuring that this knowledge is always available to meet immediate needs. In addition, GenAI significantly enhances the functionality of traditional KMS by optimizing the storage and retrieval of unstructured data (such as text) while potentially democratizing knowledge access within the organization and lowering the barriers to knowledge sharing. It can also categorize and prioritize the organization's collective knowledge, thereby improving the efficiency of knowledge management.
- *Knowledge transfer and sharing*: First, in the human-machine interaction process of GenAI as a tool, knowledge sharers can receive timely feedback and communication on new content in the field while providing knowledge, assuming the role of a knowledge recipient, thus converting knowledge sharing into knowledge exchange and enhancing knowledge flow speed, thereby improving innovation efficiency. Second, GenAI uses precise user preferences and behavior analysis to provide personalized information accurately, enhancing the efficiency of knowledge transfer direction and paths (Kumar, 2024). Finally, GenAI further weakens the contextual limitations of knowledge sharing, increasing the paths and spatiotemporal links of knowledge flow through knowledge graphs, semantic networks and data warehouses, enabling better understanding and navigation of organizational knowledge.
- *Knowledge application*: First, GenAI integrates internal and external data of an organization, assisting in analyzing, inferring and handling complex issues using methods like scenario simulation, supporting decision-makers during the complex logical thinking process, optimizing processes and driving innovation (Jarvenpaa and Klein, 2024). Second, it helps enterprises better integrate explicit knowledge both within and outside the organization, promotes the conversion of tacit knowledge to explicit knowledge within the enterprise and provides a foundation for resolving knowledge conflicts, thereby enhancing knowledge application. Finally, GenAI

generates new information and insights from current data, supporting decision-making, managing processes and building networks to facilitate knowledge application.

Building upon a systematic synthesis of extant literature, then we develop a multilevel framework examining the challenges confronting GenAI adoption in innovation ecosystems, alongside corresponding mitigation strategies.

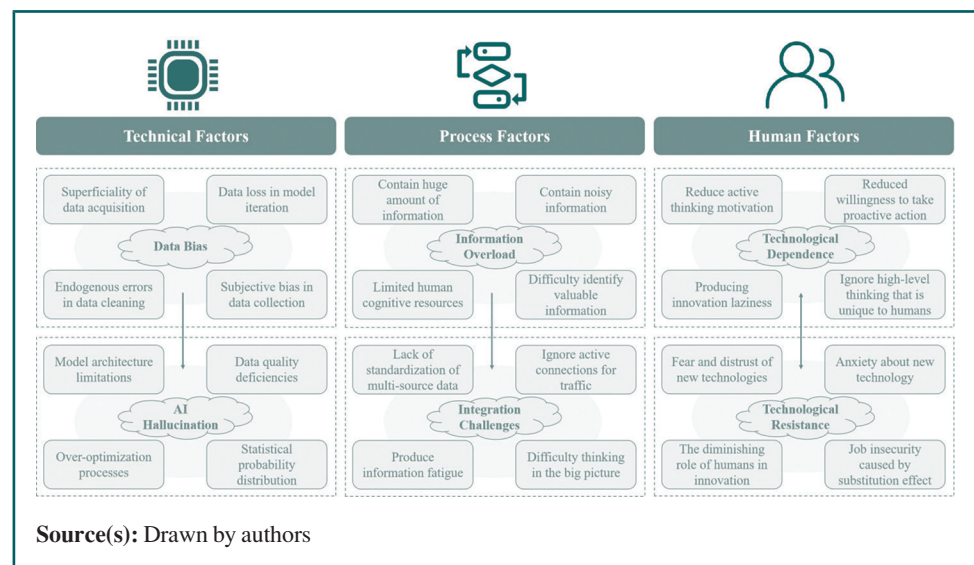
3. Dilemmas and challenges of generative artificial intelligence empowering innovation performance

GenAI has the potential to revolutionize knowledge management paradigms, thereby impacting enterprise innovation performance(Liu and Wang, 2024). Through its innovative generative capabilities, adaptability, efficiency, diversity and autonomous learning features, GenAI empowers enterprises to enhance their innovation performance. However, this process inevitably faces challenges due to factors such as the quality of raw data, the black-box nature of models and social influences. These challenges result in issues like inaccuracy, deception, misinformation and bias in the generated content. In addition, the fragmentation and dispersal of generated information can make it difficult to integrate, transfer and apply knowledge, limiting the adoption and application of GenAI in enterprise practices. As delineated in Figure 2, this paper first examines the dilemmas and challenges of GenAI in empowering innovation performance from the perspectives of the three pillars of knowledge management: technology, processes and people.

3.1 Technical level: data bias and artificial intelligence hallucination

At the technical level of knowledge management, the output of GenAI is highly dependent on the quality of its training data. In enterprises, GenAI systems often rely on historical data for training and content generation. If the input data is incomplete or contains biases – such as focusing on a specific environment or user group – the generated model may amplify these biases. Such biases can stem from limitations in historical data, data loss during model iteration, errors in probabilistic statistics and subjective errors or oversights in data collection and processing (Ma and Su, 2024). This can lead to generated content that does not accurately reflect the current market environment or user preferences, thus affecting

Figure 2 Dilemmas and challenges of GenAI empowering innovation performance



innovation decisions. For example, Amazon's attempt to develop an AI-based recruitment tool in 2014 exposed the significant impact of data bias on AI decisions. The training data, which was based on resumes submitted over the past decade, contained gender-biased keywords, causing the recruitment system to favor male candidates. The tool was eventually discontinued, revealing how historical data bias can influence AI-driven decisions.

A critical technical challenge stems from AI hallucination, a phenomenon predominantly attributed to model architecture limitations, data quality deficiencies and over-optimization processes. Defined as "the generation of outputs that deviate from input requirements or factual accuracy, wherein models fabricate non-existent content," this issue is particularly pronounced in large language models (Maleki *et al.*, 2024). The root cause lies in their reliance on statistical probability distributions, rather than authentic logical reasoning during content generation, resulting in outputs that diverge from factual reality or user intent. AI hallucination undermines enterprise innovation by inducing GenAI systems to produce erroneous assertions or misinformation, ultimately distorting innovation decision-making. Three primary manifestations characterize this phenomenon:

1. Factual misrepresentation, generation of speculative inaccuracies in factual claims or numerical data.
2. Contextual inconsistency, contradictions between generated content and input prompts, or internal inconsistencies within extended textual outputs.
3. Fabricated references, provision of non-existent citations or invalid information sources (Koehn and Knowles, 2017).

The ramifications of data bias and AI hallucination strategies may permeate organizational innovation through three critical pathways. First, enterprises risk substantial resource misallocation. Innovation initiatives predicated on flawed GenAI recommendations may squander financial and human capital, while adverse market responses could precipitate tangible economic losses (Marabelli and Newell, 2012). Second, these biases may catalyze a trust erosion within decision-making ecosystems. Uncritical reliance on GenAI-generated proposals without rigorous verification mechanisms could yield suboptimal outcomes, thereby eroding stakeholder confidence in the technology's reliability and constraining its innovative applications. Third, organizations confront prohibitive rectification costs. Mitigating AI hallucination necessitates substantial computational resource allocation and human oversight investments. Notably, current technical frameworks cannot guarantee complete elimination of hallucinatory outputs.

3.2 Process level: information overload and integration challenges

At the process level of knowledge management, although GenAI can significantly enhance enterprises' information processing capabilities, it also introduces challenges such as information overload and integration difficulties. The phenomenon of information overload is particularly evident in the application of GenAI (Kim, 2023). While GenAI generates more knowledge for enterprises, the difficulty of using this knowledge for innovation also increases. Enterprises must integrate the knowledge generated by GenAI with existing business knowledge, which first requires recognizing the value of the generated knowledge (Raisch and Krakowski, 2021). The innovative content produced by GenAI is rich in knowledge, but it also contains biases, noise and anomalies. Given decision-makers' limited resources and bounded rationality (Taylor, 1975), evaluating and filtering out low-value information becomes challenging, raising the cognitive cost of recognizing valuable knowledge during the innovation process. For instance, IBM's Watson highlights both the benefits and challenges of integrating massive data sources in health care. Watson's ability to process large volumes of medical data, such as clinical documents and patient records, is remarkable. However, it often generates excessive non-essential information, increasing

the cognitive load on doctors and potentially causing critical diagnostic insights to be overlooked. In addition, Watson faces challenges in integrating data from different medical institutions due to varying standards and formats, which can skew its recommendations.

Integration difficulties arise from the complexity of combining endogenous and external information. Differences in data formats, statistical metrics, subjective biases across information sources and variations in the use of GenAI tools by different entities make integration a complex task. Relying solely on GenAI for information transfer can create silos within organizations, hindering effective knowledge flow and sharing between individuals and departments. While the cost of obtaining information is decreasing, the cost of transforming that information into useful knowledge and applying it is increasing. Limited resources and capabilities can lead enterprises into a “technology trap,” where they struggle to manage the volume of information generated by GenAI. This may cause them to overlook the connection between information flow and business processes, impeding their ability to use GenAI-generated knowledge to reshape business processes and models. As a result, integration difficulties restrict the depth of knowledge integration, thereby hindering enterprise innovation capacity (Grover and Kohli, 2012).

Information overload and integration difficulties can degrade the quality of innovation decisions in enterprises. Information overload causes decision-makers to experience fatigue when confronted with massive amounts of data, making it difficult to identify key information and leading to hasty decisions that undermine innovation effectiveness. Integration difficulties further hinder enterprises from viewing innovation holistically, affecting opportunities for cross-entity, multi-department collaboration and the implementation of innovation outcomes. This results in a fragmented innovation process (Hislop *et al.*, 2018).

After centuries of development, human society has established a well-structured system for the “discovery-verification-dissemination” of knowledge. If GenAI, relying on current technological approaches, becomes widely adopted, it could disrupt this human-led order. Despite generating grammatically correct content, large models often mix erroneous elements with true knowledge, making it harder for people to discern and trust the information. In addition, GenAI’s innovation capability largely depends on expanding historical data and existing patterns, which can limit the depth and originality of its ideas. Especially in scenarios requiring breakthrough thinking, the limited input of conveniently generated AI information can constrain decision-makers’ ability to think disruptively and originally, thereby restricting the quality of enterprise innovation outcomes.

3.3 Human level: technological dependence and technological resistance

At the human level of knowledge management, the application of GenAI has not only transformed information processing and decision-making methods but also resulted in dual effects: technological dependence and resistance. Due to the significant advantages of GenAI in information collection and content generation, employees gradually become accustomed to relying on it for report writing, decision-making suggestions and generating innovative content. This reliance can reduce their motivation and willingness to think and analyze independently, potentially leading to a decline in creativity and innovation capabilities (Roberts and Candi, 2024). Companies that depend on GenAI for extended periods may overlook the unique human qualities of creativity, intuition and problem-solving, ultimately diminishing the company’s innovation capacity. This phenomenon of “innovation laziness” can erode an enterprise’s competitiveness.

Alongside technological dependence is the phenomenon of technological resistance. As GenAI substitutes for low-skilled jobs, some employees may resist its adoption due to distrust of the technology, fear or lack of relevant skills. Employees might feel that their role in innovation is being diminished or even replaced, leading to psychological insecurity. This insecurity can foster collective resistance within the organization, suppressing employee

enthusiasm and commitment to innovation, with a long-term negative impact on enterprise innovation. Furthermore, technological resistance may lead to counterproductive behaviors, where employees intentionally withhold, alter or obscure information required for AI training, which fundamentally restricts the application of GenAI in the enterprise innovation process. For example, BuzzFeed's decision in 2023 to extensively use GenAI for content creation triggered strong resistance from its editorial staff. Some employees feared that AI would undermine their value and job security. Despite research showing that GenAI can enhance creativity and content quality, especially for individuals with lower creativity, over-reliance on AI may reduce the diversity of innovation. This could ultimately lead to a decline in collective innovation capabilities at the organizational level (Doshi and Hauser, 2024).

Whether due to "innovation laziness" resulting from technological dependence or reduced motivation and ability caused by technological resistance and counterproductive behavior, both phenomena decrease employees' exploration of new ideas. This undermines the fundamental drivers of enterprise innovation, creating resistance during the technological transformation process. Such resistance affects the effective implementation of GenAI technologies, hinders the optimal utilization of knowledge and negatively impacts enterprise innovation performance.

Challenges arising from technical, procedural or individual dimensions collectively contribute to the misguidance of corporate innovation. In practice, these multilevel issues frequently interact dynamically, compounding their adverse effects on the precision and efficacy of innovation decision-making. Such interdependencies ultimately constrain the capacity of GenAI to enhance organizational innovation performance. This complexity necessitates systemic mitigation strategies grounded in knowledge management frameworks, which will be methodically examined in subsequent sections to address these challenges.

4. The coping path based on the perspective of knowledge management

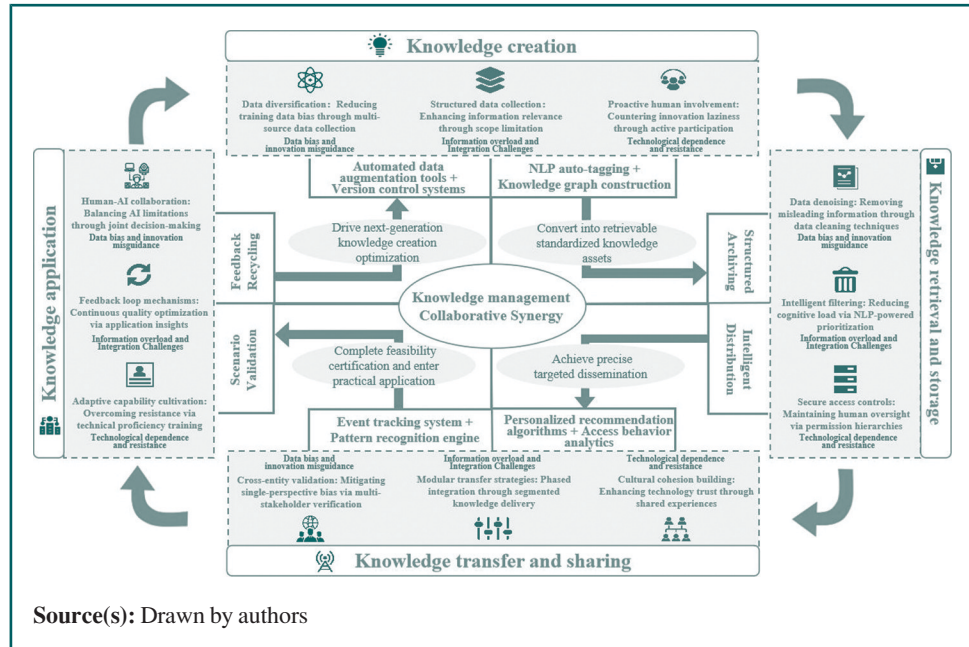
The current research provides valuable insights into the impact of GenAI on enterprise innovation performance, but the understanding of the underlying mechanisms and processes remains incomplete. Much of the discussion focuses on the benefits of GenAI for enterprise innovation performance from a technical perspective, overlooking that the value realization of GenAI is not only a technical issue but also a management issue. Successful GenAI application and value realization require not only the design, deployment and utilization of relevant technologies by enterprises but also an understanding of GenAI from a knowledge management perspective to align it with human and environmental factors within the organization. Existing research has shown that the impact of GenAI empowerment paths differs based on enterprise size and type. In this regard, this paper addresses the three aforementioned issues from the four perspectives of the knowledge management process: knowledge creation, knowledge retrieval and storage, knowledge transfer and sharing and knowledge application (shown in Figure 3). Furthermore, we propose coping strategies and paths for three categories of enterprises: startups, small and medium-sized enterprises (SMEs) and large enterprises.

4.1 Knowledge creation process

The knowledge creation process involves searching for and acquiring external information, reorganizing existing background knowledge and proposing innovative ideas and solutions. It serves as the foundation for enterprise innovation (Bhatt, 2001). To address the challenges posed by GenAI in enabling enterprise innovation during the knowledge creation process, three key aspects should be emphasized: diversification of raw data, structuring of information and human proactive involvement:

1. Emphasizing the diversification of GenAI data sources is essential. Enterprises should establish a multi-channel mechanism for collecting raw data to ensure the comprehensiveness and

Figure 3 Coping path based on the perspective of knowledge management



representativeness of data sources, thus addressing potential data biases and innovation misguidance (Schniederjans *et al.*, 2020). The diversity of input data during model training helps mitigate the risks of innovation misguidance caused by data biases. By integrating various types of data (e.g. market data, customer feedback, industry trends and policy information), enterprises can create a more comprehensive knowledge base that fosters a richer knowledge creation process. In addition, enterprises should regularly review and update data sets to adapt to rapidly changing market environments (Mariani and Dwivedi, 2024). For example, startups can leverage GenAI to quickly generate innovative ideas, reduce R&D costs and integrate GenAI tools into brainstorming sessions through small, multifunctional teams to achieve efficient innovation. SMEs can use GenAI to analyze market demand and competitive trends, providing data support for innovation directions, while establishing external collaboration mechanisms to integrate third-party knowledge and expand innovation horizons. Large enterprises can embed GenAI into R&D departments to accelerate product design and testing while creating internal innovation labs to concentrate resources and drive breakthrough projects.

- Structuring GenAI data sources is a crucial approach to addressing information overload and integration difficulties. Enterprises should establish a clear knowledge creation structure tailored to specific operational scenarios and GenAI application. This involves defining clear usage goals and standards and limiting the scope of data collection. By doing so, the relevance and applicability of GenAI are enhanced, reducing the interference of irrelevant data. In structuring data sources, it is important to balance the integration of internal and external data, avoiding over-reliance on external data that may be disconnected from the actual business context.
- In the knowledge creation process, while encouraging employees to use GenAI, enterprises must also emphasize proactive human involvement. Recent studies suggest that relying solely on AI is insufficient to drive enterprise innovation; effective collaboration between AI and humans is key to fostering innovation (Chin *et al.*, 2025). Therefore, during the knowledge creation process, enterprises should encourage employees to actively participate in the knowledge generation process, providing diverse perspectives and creative ideas to further enrich AI-generated content. Moreover, enterprises should

establish a strict verification mechanism for raw data, incorporating human oversight to prevent innovation misguidance and mitigate the pitfalls of data biases (Zahra and George, 2002). To optimize workflows, enterprises can design clear human–AI collaboration mechanisms, allowing humans and GenAI to complement each other's capabilities, maximizing innovation outcomes. For example, GenAI can specialize in data processing and pattern recognition, while employees focus on verifying innovations internally. Simultaneously, a data verification mechanism can be established to ensure the quality of the data, preventing decision-making failures caused by poor data.

4.2 Knowledge retrieval and storage process

The knowledge retrieval and storage process is crucial for creating and maintaining a knowledge repository, helping enterprises efficiently manage and use information resources to facilitate knowledge extraction and retention. To address the challenges, three aspects should be emphasized: precise data management, advanced filtering and categorization and secure knowledge repository architecture:

1. The key to addressing data biases and misguidance in the knowledge retrieval and storage process lies in precise data management. Data denoising and filtering act as gatekeepers by cleaning and removing redundant, erroneous and misleading information, ensuring that the outputs of GenAI are based on reliable, up-to-date and accurate data. Metadata management, which records key information such as data sources, creation times and usage restrictions, enables traceability for both source and generated datasets used by GenAI. Enterprises should also establish data quality control standards and governance mechanisms that address data ethics, ensuring systematic procedures for updating, auditing, verifying and evaluating databases (Becerra-Fernandez and Sabherwal, 2014). For startups, lightweight knowledge repositories can be built using cloud service platforms and GenAI tools, optimizing search engine functions to ensure efficient data access. SMEs can establish KMS integrating structured and unstructured data, using GenAI to enhance knowledge tagging and automatic categorization. Large enterprises can develop proprietary knowledge graphs to integrate data from different departments while regularly updating and cleaning knowledge repositories to avoid information overload.
2. In the data retrieval process, intelligent retrieval systems should be introduced, using advanced filtering, categorization and organization to help users quickly search, identify and handle high-priority information. By using NLP technologies and machine learning algorithms, enterprises can enhance knowledge usability, reduce decision fatigue from information overload and address knowledge integration challenges (Andreeva and Kianto, 2012). In addition, incorporating explanatory functions for generated content during knowledge retrieval can display related data sources for the results, helping employees understand the basis of the generated outcomes. This, in turn, facilitates more efficient information filtering, integration and utilization.
3. To mitigate issues of technological dependence and resistance during the application of GenAI, it is crucial to focus on the security of both source and generated databases. By implementing hierarchical access controls for the database and establishing transparent access logging and review systems, enterprises can prevent the misuse of GenAI by employees and ensure that limited access allows employees to retain necessary decision-making capabilities. Transparent usage behavior records can also reduce counterproductive behaviors resulting from employee resistance to technology. To foster team communication, enterprises can use interactive platforms to share GenAI usage methods and experiences, enhancing collaboration efficiency and building trust in technology among employees. Although these application schemes bring new challenges, they also offer opportunities for cognitive growth and enhanced teamwork, helping organizations advance in the knowledge creation process.

4.3 Knowledge transfer and sharing process

The knowledge transfer and sharing process is critical for ensuring the efficient dissemination and use of internal and external knowledge within an enterprise, driving innovation and collaboration. This process serves as both the source of data required for GenAI and a reflection of GenAI's functional utility. The challenges in knowledge transfer and sharing can be addressed by enriching and validating cross-entity knowledge, implementing knowledge transfer strategies and fostering organizational culture development:

- To address data biases and innovation misguidance associated with GenAI usage, the value of knowledge transfer and sharing lies in the richness of content during the transfer process and cross-entity knowledge validation. As mentioned earlier, diversified data sources help reduce biases. During knowledge transfer, the flow of knowledge across different departments, teams and individuals prevents knowledge silos. The widespread dissemination of knowledge promotes the aggregation of knowledge from various entities, particularly the externalization of tacit knowledge, further enriching the source data. Cross-entity knowledge transfer also facilitates knowledge validation from different perspectives, allowing for the review and refinement of generated results and minimizing biases and misguidance caused by a single viewpoint. For example, startups can use GenAI to generate visual reports and concise knowledge summaries to improve communication efficiency and share team knowledge quickly through collaboration tools. SMEs can leverage GenAI-supported internal training platforms to boost employees' awareness of knowledge sharing while promoting cross-departmental collaboration and creating incentive mechanisms for knowledge sharing. Large enterprises can develop cross-department knowledge-sharing platforms to reduce silos and use GenAI to generate multilingual reports that facilitate global team collaboration.
- The root cause of information overload and integration difficulties lies in inefficient knowledge transfer processes and unclear sharing and collaboration models. Enterprises should establish open knowledge-sharing platforms that promote cross-entity knowledge flow while developing effective knowledge transfer strategies. When dealing with large amounts of AI-generated content, enterprises should implement modular knowledge transfer strategies tailored to different stages of the innovation process, ensuring that the information volume in each unit remains manageable. This stepwise transfer helps mitigate information overload, ensures continuity in knowledge integration and reinforces the role of GenAI as a tool that supports human-driven enterprise innovation.
- Knowledge, as a carrier of culture, plays a role in enhancing organizational cultural consistency and fostering cohesion through transfer and sharing. By developing organizational culture through knowledge sharing, enterprises can cultivate innovative models of human-machine collaboration, continuously improving employees' understanding of and engagement with GenAI, including its limitations and risks. This helps reduce technology-related difficulties arising from human factors. Moreover, sharing experiences and challenges in using GenAI tools across different entities builds trust and acceptance of the technology, reduces resistance to new technologies and promotes the sharing of best practices in GenAI usage, thereby enhancing GenAI's potential to drive enterprise innovation (Bollinger and Smith, 2001). To improve adaptability to new technologies, enterprises should regularly hold training sessions and workshops, focusing on enhancing employees' technical skills, particularly in prompt engineering and iterative prompt application.

4.4 Knowledge application process

In knowledge management, the knowledge application process is the ultimate goal and the most valuable aspect. It involves effectively applying acquired, stored, transferred and shared knowledge to practical work, thereby driving enterprise innovation, optimizing decision-making and enhancing competitive advantage (Shujahat *et al.*, 2019). To address

the existing challenges in the knowledge application process, three key aspects can be emphasized: innovation decision-making mechanisms, feedback and support mechanisms and cultivating adaptability to new technologies (Wadhwa and Madaan, 2007):

1. In the era of GenAI, enterprises must establish sound innovation decision-making mechanisms, leveraging the advantages of GenAI while reinforcing the primary role of human-led innovation (Bharadwaj *et al.*, 2013). The key lies in establishing a clear human-AI division of labor and collaboration mechanism for innovation. Given the inherent limitations of GenAI in contextual knowledge, relational knowledge and generating dynamic, complex knowledge, the knowledge application process requires not only GenAI outputs and recommendations but also the integration of human higher-order thinking and judgment, especially for complex problems. Human-AI collaboration can lead to more breakthrough innovation outcomes (Bogers *et al.*, 2018). For startups, GenAI can be applied in practical business scenarios (e.g. customer support and marketing), enabling quick validation of knowledge application effectiveness through agile development methods. SMEs can integrate GenAI outcomes into core business processes (e.g. production optimization) and establish knowledge application evaluation systems to ensure practical effectiveness. Large enterprises can optimize decision support systems with GenAI, improving strategic decision-making accuracy and evaluate the long-term business value of knowledge through knowledge management tools, ensuring the sustainability of innovation outcomes.
2. In applying GenAI to enterprise innovation, knowledge application is not a one-way process; it requires the establishment of feedback and support mechanisms. Regular feedback loops should be established during the knowledge application process to assess and identify biases, noise and anomalies in AI-generated results (Curado and Bontis, 2006). Innovation and technical teams can use this feedback to continuously adjust AI usage, improve algorithms, reduce data biases and innovation misguidance and optimize outputs, thus avoiding information overload and integration difficulties during application. In addition, feedback from the application process should be integrated into the enterprise knowledge repository, promoting iterative updates and recycling of knowledge.
3. The innovative potential of GenAI depends fundamentally on users' attitudes and perceptions, which calls for strengthening their adaptability to new technologies. Adaptability is reflected both in technical proficiency and in a rational understanding of technology. Enterprises can enhance employees' technical capabilities through specialized training, diverse learning resources (e.g. online courses, workshops and practical projects) and by focusing on prompt engineering and iterative prompting, which improves efficiency in using new technologies. On the other hand, adaptability also requires a clear understanding of GenAI's capabilities and limitations (Yuwono *et al.*, 2024). Enterprises should encourage employees to maintain critical thinking, understand ethical and compliance issues (e.g. intellectual property and data privacy) and reduce anxiety brought on by rapid technological development, avoiding blind reliance on GenAI (Snyman and Kruger, 2004). In addition, enterprises should encourage employees to maintain critical thinking during use, recognizing the capabilities and limitations of generative AI in a rational manner. In particular, during the innovation application process, it is essential to understand ethical and compliance issues (such as intellectual property and data privacy), reduce anxiety caused by the rapid development of new technologies and avoid blind reliance on GenAI.

4.5 Four-process collaborative synergy

In the aforementioned four knowledge management processes (knowledge creation, knowledge retrieval and storage, knowledge transfer and sharing, knowledge application), the value of GenAI is not only reflected in the efficiency improvement and quality optimization of each process but also in the overall effect created by the dynamic interaction and collaborative synergy between the four process. When enterprises can

strategically plan and link these four process, allowing GenAI to enable the cyclical and iterative flow of data and knowledge across the entire process, its impact on driving innovation performance will be greatly amplified.

First, there is a significant feedback loop between knowledge creation and knowledge retrieval and storage. If enterprises can use GenAI to automatically extract, archive and label new data, they will continuously enrich their internal knowledge base. These constantly updated knowledge repositories will then become the source material for the next round of knowledge creation, allowing GenAI to mine and combine more diverse information, fostering the continuous incubation of innovative ideas. Second, knowledge transfer and sharing not only accelerates the diffusion of knowledge generated in the previous two process (creation and storage) throughout the organization, but also, through cross-departmental and cross-team interactions, stimulates the generation of more new ideas, providing inspiration and insights for the next round of knowledge creation (Chin *et al.*, 2021). At the same time, during the sharing process, GenAI can help identify potential stakeholders or expert networks, accelerating the dissemination of high-quality knowledge and improving its usability through personalized recommendations and collaborative tools. Finally, the relationship between the knowledge application process and knowledge transfer and sharing is often complementary and mutually reinforcing. The experiences and results validated and solidified during the application process will, in turn, enhance the sharing process' ability to distinguish between "effective knowledge" and "ineffective knowledge," helping the organization filter and iterate its knowledge. This creates a positive feedback loop. In this process, GenAI not only provides real-time insights and decision support but also structures and recognizes patterns in practical data, helping enterprises summarize replicable and scalable successful experiences and drive the diffusion and reapplication of innovation outcomes.

In conclusion, by linking the four process into a closed loop and leveraging GenAI's advantages in data analysis, content generation and cross-scenario adaptation, enterprises can maintain high levels of innovation input and output throughout the dynamic flow and continuous evolution of knowledge. At the same time, the human-AI collaboration model will be continuously optimized during this process, realizing the synergistic enhancement of each process and the overall collaborative effect, thereby maximizing the potential of GenAI in driving enterprise innovation performance.

5. Conclusion and implications

5.1 Conclusion

Traditional AI management theories (such as the resource-based view and dynamic capabilities theory) tend to focus on static resource accumulation and the dynamic shaping of organizational capabilities when explaining how enterprises use AI technology to gain competitive advantage. These theories overlook the unique advantages of GenAI in real-time learning, content generation and self-iteration. This study elaborates on the breakthroughs of GenAI in areas like dynamic knowledge generation, multimodal data fusion and deep semantic understanding, emphasizing the fundamental differences between GenAI and traditional AI in terms of technical performance and application scenarios. We further explain how these technical characteristics can significantly enhance enterprise innovation efficiency and effectiveness from a knowledge management perspective, thus injecting new perspectives into existing theories.

While the technology acceptance model (TAM) mainly focuses on individual users' attitudes, perceived ease of use and perceived usefulness when using new technology, it struggles to fully explain the multiple impacts of emerging technologies like GenAI, which have strong generative characteristics, on organizational knowledge integration, content production and decision-making processes. This study further discusses how GenAI can change enterprise knowledge processes and decision-making paths, presenting new

opportunities and challenges in knowledge retrieval, filtering, storage and innovation application. We also propose corresponding management mechanisms and implementation strategies. By comparing the performance of GenAI in knowledge cycles and collaborative innovation, we highlight how it breaks through the limitations of traditional TAM, influencing not only individual acceptance but also forming a new paradigm of human–AI collaboration in knowledge creation and management at the organizational level.

Compared to traditional AI applications that often focus on automation and decision support, GenAI's core features of knowledge generation and content creation stand out. This study, based on the four major processes of knowledge management (creation, retrieval and storage, transfer and sharing, application), explores the unique value of GenAI in each link, further emphasizing the innovation of the paper: GenAI is not just a simple replacement or complement to technological tools but forms an entire value cycle from “data-information-knowledge-insight.” This cycle not only significantly accelerates the pace of innovation but also fosters new collaborative models, enabling human–AI collaboration to play an unprecedented role in enterprise innovation.

Traditional AI management research often evaluates technological feasibility and application performance from a static or single-point perspective, neglecting the dynamic management challenges brought about by generative technologies' continuous iteration and learning. This study further discusses feasible pathways for the long-term application of GenAI in enterprise cross-department collaboration, cross-industry knowledge sharing and ecosystem construction and demonstrates its potential to reshape future organizational models. From this perspective, the paper better highlights the prospects of GenAI in improving knowledge management and innovation performance, providing a broader imagination for future research.

In conclusion, this study not only breaks through the limitations of traditional AI management theories and the TAM in explaining human–AI collaboration and enterprise innovation but also provides a new framework for constructing knowledge management and innovation systems centered around GenAI.

5.2 Research implications

This study makes several contributions to the emerging cross-disciplinary field of GenAI and innovation management, as well as to the knowledge management literature. By exploring the relationship between GenAI and enterprise innovation performance from a knowledge management perspective, this study introduces new technologies, insights and methods for enterprise innovation management in the AI era, while also offering new missions, scenarios and applications for knowledge management in the AI era.

First, GenAI, as an emerging technology capable of generating innovative content, holds immense potential for enhancing innovation performance. However, its use inevitably introduces new challenges. The intelligence of GenAI comes from training and learning processes based on machine learning. Due to the limitations of training data, and natural or technical barriers, GenAI cannot replicate explicit and tacit human knowledge. This can lead to data biases and innovation misguidance. As the cost of acquiring information decreases, the cost of converting information into useful knowledge and applying that knowledge rises, leading to issues such as information overload and integration difficulties. Sociotechnical theory emphasizes that both technological and social factors jointly influence enterprise innovation; as the primary users of technology, people's technological dependence and resistance also present challenges.

Second, previous discussions on the relationship between GenAI and enterprise innovation performance have mainly been grounded in the resource-based theory, dynamic capability theory and TAMs. Related research, from a knowledge management perspective, has primarily explored GenAI's enabling mechanisms without addressing the challenges. This study proposes a framework, from a knowledge management viewpoint, to address these challenges, breaking it

down into four key process in the knowledge management process: knowledge creation, knowledge retrieval and storage, knowledge transfer and sharing and knowledge application. It also discusses how GenAI can be used at each process to enhance enterprise innovation performance, clarifying the central role of knowledge management in enterprise innovation and expanding the application of GenAI. The framework also provides practical strategies for enterprises to tackle the challenges of the AI era and fully leverage GenAI's positive effects.

Third, this study supports the notion that relying solely on AI is insufficient to drive enterprise innovation. Effective integration, validation and application of knowledge generated by GenAI can only be achieved when a balance and collaboration between GenAI tools and human input are established during the innovation process. Although there are concerns that AI could replace routine human tasks, for complex tasks – especially those involving innovation – humans are still needed to construct meaning. Therefore, a key part of using GenAI in enterprise settings is transforming the human element. The study's findings indicate that in the process of GenAI empowering enterprise innovation, only when users possess proactive awareness, critical thinking about technology and technological proficiency, can enterprises address the negative impacts of GenAI on innovation management. This further emphasizes the role of humans as primary agents in enterprise innovation and the instrumental role of new technologies.

5.3 Limitations and future directions

This study has two main limitations that may affect the generalizability and validity of its conclusions. First, the conclusions are based on a review of existing literature and a summary of practice. Therefore, the generalizability and robustness of these conclusions need further verification through case studies or empirical research. Second, while this study explores the role of knowledge management in the relationship between GenAI and enterprise innovation performance, the underlying mechanisms and influence pathways require further examination, particularly in distinguishing the distinct roles of technological and social factors. Given the rapid advancements and expanding application of GenAI technology, future research could consider the following two directions.

First, the current framework does not take into account contextual factors such as firm characteristics, business environment and policy environment. Future research could integrate internal and external variables to build a more comprehensive theoretical model. Second, as GenAI technology evolves rapidly and its application domains continue to expand, future research could explore the differentiated impact of GenAI on enterprise innovation performance based on different human–AI collaboration models (e.g. process-based division of labor, AI as a manager, AI as a tool). This could also include assessing the multiple roles of knowledge management within these models.

Furthermore, to enhance the practical value of this study, future research could focus on new digital platforms such as data platforms, business platforms and microservices architectures, beyond the traditional knowledge management platform. These platforms could integrate data, processes and resources across different business systems or departments, creating a unified and scalable environment for GenAI in knowledge acquisition, governance and application. Finally, we look forward to more scholars actively participating in interdisciplinary research on AI and innovation management, collectively advancing the application framework of new technologies in management practices and shaping the future research agenda for AI technology management.

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